



Name: _____

Class: _____

Date: _____

Mr. Rawson • IB DP Physics

Simple Harmonic Motion 5 – Pendulum Lab — Measuring g

Practical

As a class, you are about to measure the strength of gravity using nothing but a length of string, a mass, and a stopwatch. Each pair takes one length; the whole class's results together make the graph.

Key concept

For a simple pendulum swinging through a small angle,

$$T = 2\pi\sqrt{\frac{l}{g}}$$

Squaring both sides straightens the relationship, exactly as you did for the spring: $T^2 = \frac{4\pi^2}{g}l$. A graph of T^2 against l is a straight line through the origin, with gradient $4\pi^2/g$ — so the gradient of *today's* graph hands you a value for g .

Your length

Seven pairs, seven lengths, 0.5 m apart. Find your pair in the table and circle your length.

Pair 1	Pair 2	Pair 3	Pair 4	Pair 5	Pair 6	Pair 7
0.5 m	1.0 m	1.5 m	2.0 m	2.5 m	3.0 m	3.5 m

Method

1. Set up your pendulum at your assigned length l , measured from the pivot to the **centre of the bob**. **Check:** the string itself should measure a little less than l — the bob's own radius makes up the rest.
2. Pull the bob to one side by about 10–15° and release from rest, with no push.
3. Time **10 complete oscillations**, starting your count at zero as the bob is released.
4. Repeat for a total of **5 timings** at your one length.

Your data

Length l / m	Time for 10 oscillations / s					Mean t / s
	1	2	3	4	5	

Your period, $T = \text{mean } t \div 10 =$ _____ s $T^2 =$ _____ s²

Report your l and your T^2 to the board once your five timings are done.

The whole class's data

Copy every pair's result from the board — seven points make a graph, one point does not.

Length l / m	0.5	1.0	1.5	2.0	2.5	3.0	3.5
T^2 / s^2							

Processing

1) **Plot** a graph of T^2 (vertical) against l (horizontal) using the class's seven points, and draw the best straight line through them.

[3]

2) **Determine** the gradient of your line, and use it to calculate g .

[4]

My gradient = _____ s²/m

$$g = \frac{4\pi^2}{\text{gradient}} = \text{_____ m/s}^2$$

3) **Compare** your calculated g with the accepted value, 9.81 m/s². Give a percentage difference.

[2]

Key concept

$$\text{percentage difference} = \frac{|\text{your value} - \text{accepted value}|}{\text{accepted value}} \times 100\%$$

The upright bars mean *take the size of the difference and drop the sign*, so a value that comes out high and one that comes out low by the same amount give the same answer.

Percentage difference: _____ %

4) **Explain** why plotting T^2 against l , rather than T against l , lets a straight line hand you a value for g .

[2]

Before Tuesday

Homework

Finish this worksheet — the graph, the gradient, your value of g , the percentage difference, and the last question — due at the start of Tuesday's lesson.